

1. Administrative & Study Identification

Full Study Title: Novel Combinations in Participants With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Adenocarcinoma **Official Title:** An Open-Label, Multi-Drug, Multi-Centre, Phase II Study to Evaluate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Immunogenicity of Novel Combinations in Participants With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Adenocarcinoma **Short Title/Acronym:** GEMINI-Gastric **Protocol Number:** D7986C00001 **NCT Number:** NCT05702229 **Posting ID:** 564919878184412695 **Trial Status:** Recruiting **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Novel combinations (volrustomig, rilvegostomig, AZD0901, or AZD7789) plus Chemotherapy (including fluorouracil-based regimens like FOLFOX or XELOX) **Phase of Development:** Phase II **Medical Monitor:** AstraZeneca Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the objective response rate (ORR) and progression-free survival (PFS) at 6 months by investigator assessment per RECIST v1.1. **Secondary Objectives:** To evaluate safety, tolerability, duration of response (DoR), overall survival (OS), pharmacokinetics (PK), and immunogenicity of the novel combination therapies.

2.2 Background & Rationale To address the need for improved outcomes in patients with locally advanced unresectable or metastatic gastric or gastroesophageal junction (GEJ) adenocarcinoma who are HER2-negative. While platinum- and fluoropyrimidine-based doublet chemotherapy remains the standard of care, recent data suggests immuno-oncology (IO) agents combined with chemotherapy can improve overall survival. This master protocol assesses multiple novel bispecific antibodies and antibody-drug conjugates (ADCs) in combination with chemotherapy to determine the most effective and safe regimens for this patient population.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Master Protocol / Platform Study) **Control Type:** Parallel Assignment (across multiple substudies) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 240 participants (assigned across 6 substudies) **Number of Sites:** Global multi-center (including US, Europe, and Asia [UK, Spain, Taiwan, South Korea, Japan, China]) **Estimated Study Start:** 16-Jan-2023 **Estimated Primary Completion:** 30-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of signing the ICF. 2. Body weight > 35 kg. 3. Previously untreated for locally advanced unresectable or metastatic gastric or GEJ adenocarcinoma, with measurable target disease based on RECIST 1.1 and ECOG PS 0 or 1. 4. Life expectancy of at least 12 weeks. 5. Adequate organ and bone marrow function. 6. HER2-negative status. Has central lab confirmed Claudin18.2 status at screening from archival tumour collected within the past 24 months or from a fresh biopsy when Substudy 3, Substudy 4 or Substudy 6 is open for recruitment.

4.2 Exclusion Criteria 1. Participants with HER2-positive (3+ by IHC, or 2+ by IHC and positive by in situ hybridisation) or indeterminate gastric or GEJ carcinoma. 2. Untreated or progressive CNS metastatic disease, any leptomeningeal disease, or cord compression. 3. Participants with ascites which cannot be controlled with appropriate interventions. 4. Active infectious diseases, including tuberculosis, HIV infection, or hepatitis A/B/C. 5. Uncontrolled intercurrent illness. 6. Active or prior documented autoimmune or inflammatory disorders requiring systemic treatment with steroids or other immunosuppressive treatment, or history of another primary malignancy. 7. Previous treatment with an immune-oncology agent. 8. Previous treatment with any modalities of Claudin18.2 targeted therapy or MMAE exposure (when Substudy 3, Substudy 4, or Substudy 6 is open for recruitment).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administration of novel agents (e.g., MEDI5752, AZD2936) assigned via substudies, in combination with the investigator's choice of standard chemotherapy regimens (such as fluorouracil-based FOLFOX or XELOX). **Route of Administration:** Intravenous (IV) infusions. **Duration of Treatment:** Maximum treatment period of up to 24 months, or until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard palliative and supportive care medications to manage symptoms and chemotherapy-related toxicities. **Prohibited:** Previous treatment with an immune-oncology agent, previous treatment with any modalities of Claudin18.2 targeted therapy or MMAE exposure (for

corresponding substudies), and concurrent immunosuppressive therapies.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, RECIST 1.1 Baseline Scans, Biomarker (HER2/Claudin18.2) Testing | | **Baseline** | Day 0 | Substudy Assignment, First Dose of Investigational Product + Chemotherapy | | **Interim** | Every Cycle | Safety Labs, PK/Immunogenicity Sampling, Efficacy Assessment (Tumor Imaging per RECIST 1.1) | | **End of Study** | Up to 24 Months | Final Restaging Scans, Exit Interview, IP Accountability, Transition to Survival Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR) and Progression-Free Survival at 6 months (PFS6). **Measurement Method:** ORR (confirmed complete response or confirmed partial response) and PFS6 (alive and progression-free at 6 months) via investigator assessment evaluated according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1.

7.2 Secondary & Exploratory Endpoints 1. Progression-Free Survival (PFS) per RECIST 1.1 as assessed by the Investigator. 2. Duration of Response (DoR) and Overall Survival (OS). 3. Other safety related endpoints, tolerability, PK, and immunogenicity profiles of the novel combinations.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during regular dosing visits, focusing on immune-related adverse events (irAEs) and chemotherapy toxicities. **Serious Adverse Events (SAEs):** Standard reporting timeline within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** A Safety Review Committee (SRC) evaluates data to confirm the recommended dose for study interventions in each corresponding substudy/cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy Evaluable Set (approximately 40 evaluable participants of the confirmed recommended dose by SRC per substudy) and Safety Set for AE/SAE evaluations. **Sample Size Justification:** \$N \approx 240\$ sample size has been selected to provide adequate statistical precision for the co-primary endpoints of ORR and PFS at 6 months across the 6 planned

substudies. **Handling of Missing Data:** Handled according to AstraZeneca's standard methodologies for missing RECIST evaluations and censored survival data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Ongoing onsite and remote monitoring visits to ensure site protocol compliance and data integrity. **Audit Trail:** Robust data management with eCRF locking and query resolution via standard data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D7986C00001 **Registry Number:** NCT05702229 **Posting ID:** 564919878184412695 **Trial Status:** Recruiting **Sponsor/Lead**

Organization: AstraZeneca **Study Title:** Novel Combinations in Participants With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Adenocarcinoma **Official Regulatory Title:** An Open-Label, Multi-Drug, Multi-Centre, Phase II Study to Evaluate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Immunogenicity of Novel Combinations in Participants With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Adenocarcinoma

12. Clinical Framework (Gastric Cancer Specific)

Primary Condition: Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction (GEJ) Adenocarcinoma (Gastric Cancer) **Diagnosis Threshold:** Histologically/cytologically confirmed HER2-negative advanced disease **Medical Methodology:** Master Protocol / Platform Study of Immuno-Oncology Agents plus Chemotherapy (FOLFOX or XELOX) **Optimization Target:** Improvement of Objective Response Rate (ORR) and 6-month Progression-Free Survival (PFS6)

13. Study Procedures & Methodology

Management Pathway: Novel IO + CTx Master Protocol substudy assignment **Education Protocol (HCP):** Site initiation and protocol training for RECIST v1.1 evaluation, bispecific antibody management, and irAE handling **Education Protocol (Patient):** Informed consent process, treatment schedule education, and toxicity awareness **Quality Audit Mechanism:** Safety Review Committee (SRC) evaluations, source data verification, and central laboratory verification for biomarkers

14. Observation & Follow-Up Schedule

Total Duration: Up to 24 months of treatment + subsequent survival follow-up **Onsite Visit Cadence:** Aligned with required chemotherapy cycles (e.g., Q2W or Q3W) and tumor imaging schedules **Remote Monitoring Frequency:** Intermittent survival follow-up post-treatment discontinuation **Checklist Review Interval:** Ongoing AE/SAE assessment prior to the administration of each treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						